Team name	Pray First
Project title	AcademiaFlow: Unified Student Lifecycle Workspace for Course Planning, Lecture Mastery, Accountable Groupwork, Internships, and Thesis SLA Tracking
Selected case	Case 1: Smarter Campus and Learning
Members	Allfridhoo Prima Anjaynata Saragi (Leader), Mikhael Yuli Ananda Elvan Permana, Nathanael Christian Putra

Abstract

1. Problem. Across their university lifecycle, undergraduates face four connected bottlenecks caused by rigid legacy portals and scattered chat groups: (a) during course registration (KRS/FRS), students lack syllabus previews, face server slowdowns, and get separated from their study circle; (b) in lectures and group projects (Tugas Besar), fast-paced slides hinder comprehension while lecturers cannot track individual progress or prevent free-riders; (c) in semesters 5–7, scholarship and MoU-internship posters get buried in chats, delaying 20-SKS conversion and alumni networking; and (d) during thesis (Tugas Akhir) supervision, unstated advisor quotas, messaging anxiety, and extended advisor inactivity stall graduation.

2. Intended Users. First- through final-year computing and engineering undergraduates, course lecturers supervising team projects, and program coordinators (Kaprodi/Dosen Wali) overseeing credit conversions and thesis advisor workloads.

3. Supporting Observations or Evidence. From multi-campus audits and interviews across ITB, Telkom University Surabaya, and UNESA [1], students identified stalled thesis revisions from overloaded advisors as their top graduation bottleneck, alongside untransparent quotas and WhatsApp messaging anxiety—aligning with Indonesian higher-education studies [2]. Informants also confirmed frequent free-riding in lecturer-assigned groups tracked only via manual sheets [1, 3], buried opportunity posters in student association chats [1], and slow manual 20-SKS MBKM credit consultations [4].

4. Proposed Solution and Core User Flow. AcademiaFlow is a semester-adaptive web workspace using one-click KRS PDF import: (1) Smart KRS & Circle Sync: previews 14-week syllabi, practicum schedules, and lecturer styles while simulating conflict-free parallel classes with live quota indicators; (2) StudyPulse & Accountable Groupwork: students drop lecture audio and slide PDFs to generate slide-synced transcripts, Socratic explanations, FSRS active-recall quizzes, and shared notes, while team boards auto-intersect free hours, trigger relay handover alerts, stream contribution logs to lecturers, and support free-rider escalation; (3) Adaptive Opportunity & Alumni Hub (Semesters 5–7): surfaces filtered scholarships, MoU internships with 20-SKS conversion simulators, activity-point (TAK/SKKM) trackers, and opt-in alumni contacts; and (4) Thesis SLA & In-App Supervision: provides semantic advisor matching, standardized consultation tickets with one-click logbook approval, and 21-day inactivity alerts recommending replacement advisors. The initial version targets computing cohorts.

5. How It Differs from Existing Alternatives. Legacy portals (SIX, iGracias, SIAKAD) only record registration and final grades, whereas generic chat, note, and kanban apps lack KRS schedule intersection, slide-synced audio active recall, lecturer contribution telemetry, MoU SKS mapping, and advisor SLA rebalancing. AcademiaFlow unifies these workflows without requiring write-access to university databases.

6. Planned Technologies. A Next.js 15 and TypeScript web app backed by Supabase (PostgreSQL, pgvector, Auth, and Realtime WebSockets for schedule rules, handover alerts, and SLA timers). Collaborative notes and quizzes use BlockNote, Yjs, and ts-fsrs, while a FastAPI service uses faster-whisper, MarkItDown, PaddleOCR, and Google LangExtract (Gemini API) to parse lecture audio, KRS/syllabus PDFs, and posters into JSON.

7. Expected Benefit and Development Plan. Expected benefit: cut exam review and group coordination time by 50%, curb free-riding, and reduce SKS consultation and logbook approval under 5 minutes in a two-week trial with 30 students. Plan: Week 1 KRS Circle Sync and StudyPulse/Groupwork Engine; Week 2 Opportunity/Alumni Hub and Thesis SLA Rebalancer; Week 3 pilot testing. Unavailable campus APIs and OCR/ASR errors are mitigated via PDF parsing and confirmation previews.

Source Acknowledgments. [1] Pray First Team, Multi-campus workflow audit and student interviews across ITB, Telkom University Surabaya, and UNESA (Nov 2026). [2] Indonesian higher-education studies on external supervisor bottlenecks and communication anxiety in undergraduate thesis completion. [3] Empirical studies on social loafing, free-riding, and individual accountability in Indonesian university group coursework. [4] National evaluation studies on 20-SKS MBKM internship credit conversion and academic advising bottlenecks.